

Geometric versus classical integrators for black-hole geodesics

Research summary

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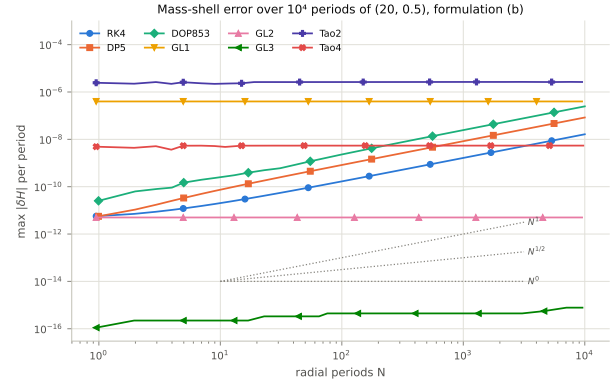
Question. Black-hole imaging traces millions of short photon paths; inspiral modelling follows a few orbits for thousands of revolutions. Which numerical integrator should each use, and why? I compared eight integrators on geodesics of the Schwarzschild metric: classical and adaptive Runge–Kutta (RK4, DP5, DOP853), Gauss–Legendre collocation of order 2, 4 and 6, and Tao’s explicit symplectic scheme. Each runs on the second-order geodesic equation and on Hamilton’s equations, and every error is measured against an exact solution built from elliptic integrals and functions, for light deflection, the photon sphere, circular and eccentric orbits, the ISCO and inclined orbits.

Findings.

- *Error growth.* Over 10^4 orbits the Runge–Kutta methods drift linearly in the mass shell (RK4: 1.6×10^{-8}) and quadratically in phase; the symplectic methods stay bounded (GL2: 5.0×10^{-12}) with linear phase error. Sixth-order Gauss–Legendre reaches round-off and follows Brouwer’s $N^{1/2}$ law.
- *Formulation.* In Hamilton’s form, energy and angular momentum are exact for every method; the second-order form loses up to 440 times in accuracy near the critical impact parameter. A method that is merely symmetric there conserves as well as a symplectic one.
- *Cost.* DOP853 is cheapest for short integrations at every accuracy. Over 1000 orbits only sixth-order Gauss–Legendre reaches a phase error of 10^{-8} (6.2×10^3 evaluations per orbit).
- *Edges of stability.* On the photon sphere accuracy buys survival only as $\ln 10/2\pi$ orbits per decade, capped near six orbits in double precision; plain summation fakes a longer life. At the ISCO the time to plunge scales as $|\varepsilon|^{-1/4}$ in the error size, and the error’s sign decides whether the orbit plunges at all.

How. The code (Python with Numba) keeps the full eight-dimensional phase space, so conserved quantities are measured, not imposed, and separates integrators from physics: they see only a vector field. Every figure and table is regenerated from a clean checkout by one command, and a repeated run reproduced every result byte for byte. A development log records what failed and why, from a mistranscribed elliptic formula caught by an independent check to an integrator parameter that fails on marginally stable orbits.

Next. The metric interface takes the full inverse metric and its derivatives, so the Kerr metric enters as a new class with no change to the integrators. This connects directly to my earlier work on light deflection in Kerr–(anti-)de Sitter spacetimes: the same framework can test numerical deflection angles against analytic results where no closed form for the whole orbit exists, and extend the stability results here to rotating black holes.



Mass-shell error over 10^4 radial periods of an eccentric orbit: linear drift for the Runge–Kutta methods, bounded error for the symplectic ones, Brouwer’s law for GL3 at round-off.